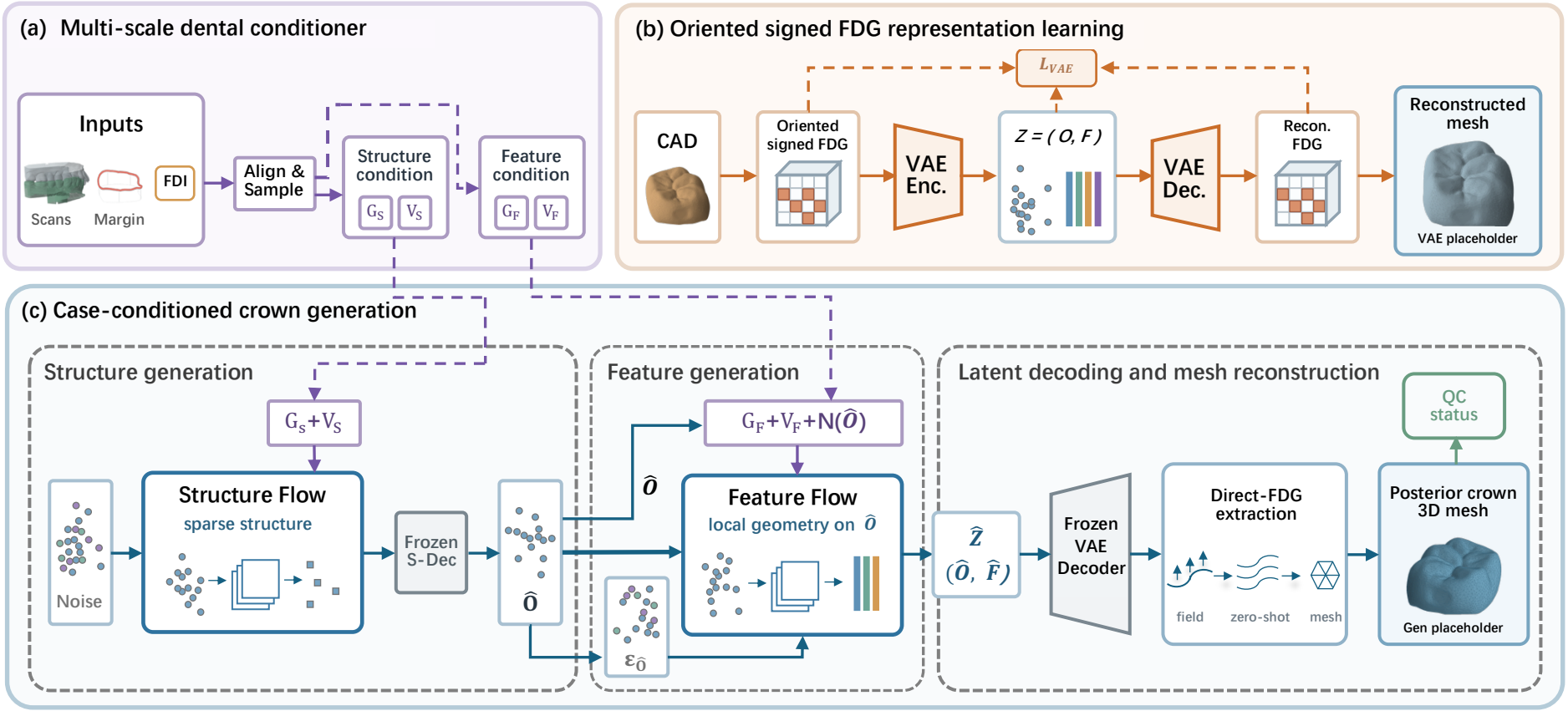

(a) Multi-scale dental conditioner
Inputs
Scans
Margin
FDI
Align & Sample
Structure condition
G_S V_S
Feature condition
G_F V_F
(b) Oriented signed FDG representation learning
L_{VAE}
CAD
Oriented signed FDG
VAE Enc.
Z = (O, F)
VAE Dec.
Recon. FDG
Reconstructed mesh
VAE placeholder
(c) Case-conditioned crown generation
Structure generation
G_s+V_S
Noise
Structure Flow
sparse structure
Frozen S-Dec
$\hat{O}$
Feature generation
G_F+V_F+N($\hat{O}$)
$\hat{O}$
Feature Flow
local geometry on $\hat{O}$
$\varepsilon_{\hat{O}}$
Latent decoding and mesh reconstruction
$\hat{Z}$
($\hat{O}$, $\hat{F}$)
Frozen VAE Decoder
Direct-FDG extraction
field
zero-shot
mesh
QC status
Posterior crown 3D mesh
Gen placeholder